

How Tos

Project: Group Management (INFIBIO)

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How to grow *Chlamydomonas* in your lab

Procuring *Chlamydomonas* strain

1. Different strain (for example 'wild type') can be procured from [Chlamydomonas resource center](#) or can be obtained from a laboratory working on that strain.

Culturing in the lab

Once the culture is received from the resource center/lab, the procedure to grow/propagate the cell culture is following:

Note: Detailed algal culture techniques are provided in the book: (i) [Algal Culturing Techniques](#) and (ii) [The Chlamydomonas Sourcebook](#).

Equipment and materials:

1. Temperature controlled (22-25 degrees Celsius) Incubator with orbital shaker and cool white fluorescent light (between 50 and 100 $\mu\text{Em}^{-2}\text{s}^{-2}$) (**Note:** Light should be controlled with a digital programmable timer so that period of light/dark cycle can be set; try to use a light source with a spectrum that is good for growing plants. In the past we were using OSRAM Fluora fluorescent tubes, but now they are not in production anymore. We now use LED fluorescent tubes for plant growth.)
2. TAP media: Protocol to make the TAP media is available online at [Chlamydomonas Resource center](#), [protocols.io](#). TAP can also be purchased from [Thermo Fisher Scientific](#), .
3. Glass conical flask (or culture flask) (for liquid culture)
4. Laminar flow hood
5. Serological pipette
6. Inoculation loop
7. Bacteriological agar (to grow cells on the solid agar surface)
8. Petridish plate (to grow cells on the solid agar surface)

Procedure to prepare liquid culture:

1. UV-sterilize the hood and clean the surface with 70% isopropanol. Bring the stock culture (solid or liquid), culture flask (previously autoclaved/ sterilized with the cap or cotton plug on), inoculation loop, serological pipette, and TAP media inside the hood while the flow hood is ON.
2. Add 20 ml of TAP media using serological pipette in culture flask of volume 100ml and close the flask with a cap or with a cotton plug.
3. Using the inoculation loop (if the stock is a solid culture), pick a single colony and carefully add it to the conical flask containing 20ml TAP media.
4. Bring the culture flask and place it on the orbital shaker in the incubator. Set the orbital shaker at 100 rpm (might be a good idea to try out other speeds if you suspect that the aeration might be insufficient).
5. Set the light/dark period to be 14h/10h for everyday.

6. Cell culture will be ready in 5-6 days following a logarithmic growth.
7. The first cultures from the solid culture are usually clumpy (i.e., a large fraction of cells will be in clumps of 2-3 cells), and therefore, we propagate the culture further.

Further propagation of the liquid cell culture:

1. Repeat the step from 1 to 2 of 'Procedure to prepare liquid culture:'.
2. Using serological pipette/sterile pipette add few hundred microlitre to the conical flask containing new fresh TAP media. (Best way is to characterize the growth rate of the culture and add appropriate concentration of cells by counting cells using counting chamber.)
3. Repeat the step 4 to 5 from 'Procedure to prepare liquid culture:'.
4. This culture will be ready in 2-3 days.

Procedure to prepare solid culture:

1. Prepare liquid culture as mentioned in the 'Procedure to prepare liquid culture:'.
2. Add 25 ml TAP medium in a glass conical flask.
3. Weigh the appropriate amount of Bacteriological agar powder for 1.5% solution in TAP (For example, 0.375 g for 25 ml TAP) and add to the conical flask containing TAP media.
4. Autoclave the mixture. After autoclave the solution will become slightly viscous.
5. When temperature of the solution reaches to 40-50 degree transfer it in the falcon tube and keep it slanted on a stand and let it cool down (eventually leading to solidification).
6. Bring the liquid culture in the laminar flow hood. Dip the inoculation loop in the liquid culture and streak the solid agar surface in the falcon tube.
7. Tighten the cap of the falcon tube and wrap a parafilm around it.
8. Store the falcon tube in presence of the cool white fluorescent light ($\sim 20\text{-}30\text{ }\mu\text{Em}^{-2}\text{s}^{-1}$). Generally, solid culture requires less light intensity.
9. This falcon tube can be stored as stock and should be remade at every 3 months.

Note:-

1. Use the cell culture from the same day (in the logarithmic phase) and time (approximately) for the reproducibility of your measurement.
2. Stock your culture as a solid culture on an agar surface in case you intend to use *Chlamydomonas* in the future. Remake the stock every three months.

BSA-treated glass

Objective: To prevent cell adhesion on the glass surface we use [Bovine Serum Albumin](#) (BSA) to coat the glass slide and cover slip.

Equipment and Material

1. BSA Powder
2. 100 ml Beaker
3. 50 ml Falcon tube

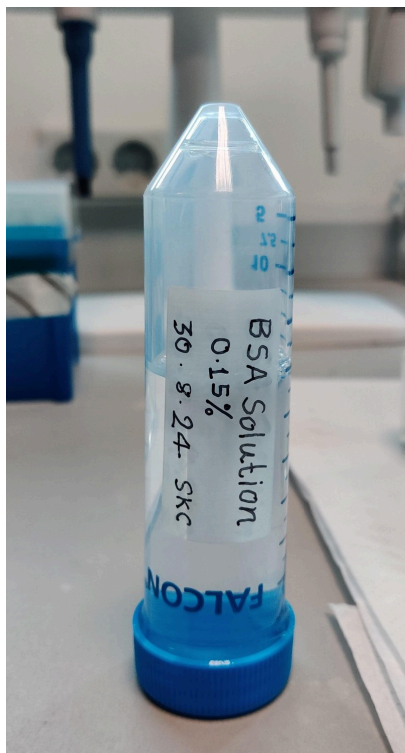
4. Glass slides and Cover slips
5. Coplin Jar (optional, instead a Petri dish or other appropriate glass container can be used)
6. H_2O_2
7. milliQ water (or dH_2O)
8. 0.2 μm syringe filter (optional)
9. Plasma chamber (optional)

Procedure:

Preparation of 0.15% BSA Solution (25 ml solution):

1. Weigh 37.5 mg of BSA powder and put them in a clean autoclaved beaker. Mix the solution well.
2. Take the solution inside the laminar flow hood and filter sterilize it using a 0.2- μm syringe filter. (**Tip:** This step can be skipped if the entire solution is used on the same day.)
3. Store the sterile solution in a 50 ml sterile falcon tube at 4 degree. Be careful to prevent the solution from contamination if intended to use the same solution for longer period of time.

 BSA_Solution.jpg



Normal Cleaning of the Glass slide and cover slip:

1. Arrange the coverslips and glass slides in a Coplin Jar and fill it with H_2O_2 . Let the glass slide and coverslip soak in H_2O_2 for around 24 hours (there is no maximum time limit).

2. Take out the glass slide and cover slips using tweezer and rinse it well with milliQ water (or dH₂O). Shake off excess water.
3. Air dry the coverslips and glass slides with compressed Nitrogen or dry air. Alternatively, you can try drying the coverslips on a hot plate, but make sure as much water as possible has been removed beforehand. (**Tips:** a) a classical way to remove water is to rinse it with ethanol or isopropanol (70%); b) remember that if you use low-grade media, they will tend to leave a solid residue onto the glassware as it dries; c) maintain the cleanliness of the slides/coverslip after this step, for example, by keeping it within a clean closed container.)

Normal_Cleaning.png



Glass surface activation by plasma treatment (Optional):

This step activates the glass surface, leading to faster binding of BSA to the glass.

1. Switch on the plasma chamber machine and make the appropriate settings in the program for 10 minutes of plasma treatment
2. Place the clean glass slides and coverslip inside the chamber, then close the door. The side facing up will be activated.
3. Run the program. (**Tips:** The machine should produce vacuum within the chamber sufficiently slowly. If you produce vacuum too fast, light objects such as coverslips can be displaced by air drafts and topple on top of each other inside the chamber)

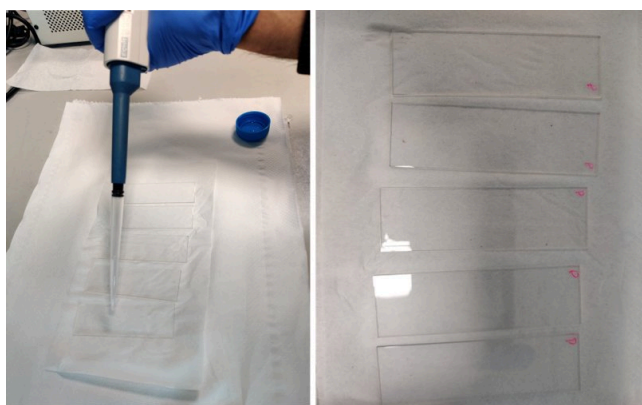
Plasma_treatment.png



Incubation of glass slide and coverslip in BSA solution:

1. Immediately after plasma treatment, place the BSA solution using a pipette onto the region of interest or onto the entire glass slides and coverslips. (**Tips:** a) Mark the slides/coverslip to identify the coated side; b) One can soak the entire glass slide and coverslip in the BSA solution, but be aware that the remaining solution may not be usable after a few days due to contamination)
2. Leave the glass slide/coverslips with BSA solution for 1 hour at room temperature. The system can also be left at 4 °C in the fridge.
3. Rinse the glass slides/coverslips with Milli-Q water and air dry. They are ready to use and can be stored in a clean container for use in the next couple of days.

 Slide_Incubation_BSA.png



Note: If a plasma chamber is unavailable, one can skip this step and incubate the glass slide and cover slip in the BSA solution at least 24 hours.

Preparation of PDMS chamber:

Making channel in PDMS:

The process of making channels in **Polydimethylsiloxane (PDMS)** material is called soft lithography. These channels are widely used in microfluidics and possess shape deformability and air permeability.

The PDMS kit contains Silicone elastomer and hardening agents which are [commercially available](#).

1. Keep the mold ready. Molds are generally made using photolithography, especially if the design's dimension is below 100 microns. However, a cheaper alternative can be used. Here, we cut a coverslip with a diamond cutter into a rectangular shape (LxWxH: 30mmX5mmX160um) to use as a mold.
2. Weigh the elastomer and hardening agent into a clean container at a 10:1 (elastomer:hardening agent) ratio, then vigorously mix the solution for 10 minutes (this time is for approximately 40-50 g of solution; adjust as needed for the amount).

3. After mixing, the solution will become milky whitish; many air bubbles will be trapped in it. To remove trapped air bubbles, degas the solution in a desiccator for almost half an hour (or until there are not many air bubbles). A few remaining air bubbles will automatically pop out when the desiccator lid is opened. After that, even if a few air bubbles remain on the surface, they can be popped out by touching with a sharp needle's tip.
4. Keep the mold in a clean Petri dish plate (which can withstand 70-90 degrees Celsius). One should use more molds with appropriate spacing to maximize the use of the PDMS solution.
5. Pour the solution into the Petri dish containing the mold of the required design and let the solution spread evenly. (**Tip:** if the mold is a very light object, it will start floating in the solution. To prevent this, for example, in the case of the piece of the coverslip, use a very small drop of water between the coverslip and the Petri dish surface, which will immediately lead to sticking due to capillary force. Let it dry and then pour the PDMS surface)
6. (**Optional**) Put the entire Petri dish on a hot plate (or in a hot air oven) at 70 degrees Celsius for 1-2 hours to cure. The time of curing and temperature can be adjusted based on the amount of PDMS solution. (**In case of unavailability of a hot plate or oven**, one can leave the solution for more than 24 hours at room temperature to cure.
7. After proper curing, the solution becomes hard (one can test this by pressing it with a toothpick or similar item).
8. Using a scalpel, carefully cut out the PDMS (usually in a square or rectangular shape around your design) and gently peel it off.
9. (**Optional**) The square/rectangular block can be cut further based on the requirement (**Tip:** Use a rubber mat on which you keep the PDMS and cut it).
10. If necessary clean the PDMS using isopropanol followed by rinsing it with dH₂O (or milliQ water).

Bonding of PDMS with glass surface:

Glass slide/coverslip is used to make the PDMS closed from the open side. Following are procedures to bond with glass surface:

1. Clean the glass slide/coverslip and PDMS properly and air dry them.
2. Put them in the Plasma chamber, keeping the binding surface facing upwards to activate the surface.
3. Immediately after the plasma treatment (not more than 2 minutes), gently bring the treated surface of the PDMS and glass slide in contact, facing each other. Place the chamber on the hotplate at around 80-100 degrees Celsius for 1-2 minutes. The binding process is quick and permanent. The PDMS device is ready for use. [**Tip:** Plasma bonding of glass slide and PDMS can be skipped if the proper and permanent bonding is not necessary. Instead, ensure the glass slide and PDMS are properly clean and smooth. Gently bring the binding surface of the glass slide and PDMS in contact. The PDMS will adhere to the glass surface and does not come off on its own unless it is taken off. In this case, removing the PDMS from the glass slide is easy to clean and reuse.]
4. (**Optional**) Inlets and outlets are usually made by punching the PDMS using a punching tool (**PDMS Chip Hole Puncher or Biopsy puncher**) and **microfluidic tubing** to insert any liquid sample. (**Tip:** For a simple chamber design, such as a circular well with a few mm diameter and depth of hundreds of microns, making an inlet and outlet chamber may not be necessary. For such a well, place the appropriate amount of liquid sample in the well in the PDMS, then adhere the glass slide from the top by simply bringing it into contact with the PDMS. **Limitation: In this process, there is a high risk of spillage from the chamber or of trapped air, depending on the amount of liquid in it.** Another simple design is a rectangular channel with an open end (serving as both inlet and outlet), in which making separate inlets and outlets is unnecessary. In this case, first adhere the PDMS to the glass surface by gently bringing it into contact. Then insert the liquid sample from one end using a pipette. (**Limitation 1:** Due to the hydrophobic nature of PDMS, it will be a bit hard to push the liquid in the PDMS channel, which can also lead to trapping of air bubbles. This problem can be resolved by coating the channel with an appropriate coating agent, such as BSA.) **Limitation 2:** In this case, due to the open end of the channel, there could be advective flow in the chamber, which can be prevented by sealing both open ends with Vaseline.)

5. **(Optional)** To prevent cell adhesion on the glass or PDMS surface, one may coat the device using an appropriate coating agent, such as a BSA solution.
6. These PDMS channels can be reused after proper cleaning.

Preparation of a thin spacer (90um) of PDMS

- **Required Equipment and Materials**

- Spin coater
- PDMS solution
- Petri dish
- Hot plate

- **Procedure**

- Prepare the PDMS solution as described in the above section ("Preparation of PDMS chamber").
- Place the 90 mm diameter Polystyrene disposable Petri dish (size can be optimized based on the requirement) on the base of the rotor (or chuck). Make sure it is fixed properly. Double-sided tape can be used to fix it.
- Pour 1ml of PDMS solution on the petridish.
- Rotate at 750rpm for 60 s. (These values may be required to optimize based on the type of spin coater)
- Immediately heat to cure at 60 degrees C for 15 min.

This provides us with a ~90 μm -thick layer of PDMS, which can be easily peeled off the petri dish. Using a biopsy puncher and a scalpel, the required shape can be cut out to form a quasi-2D chamber for studying the motility of microorganisms.